



SOBERANIA: DEVELOPING SOVEREIGN LLMS FOR BRAZILIAN PORTUGUESE

**CASE STUDY: OBSTACLES,
BREAKTHROUGHS, AND PUBLIC
SECTOR LEADERSHIP IN LOW-
RESOURCE AI**

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MARCELLUS AMADEUS

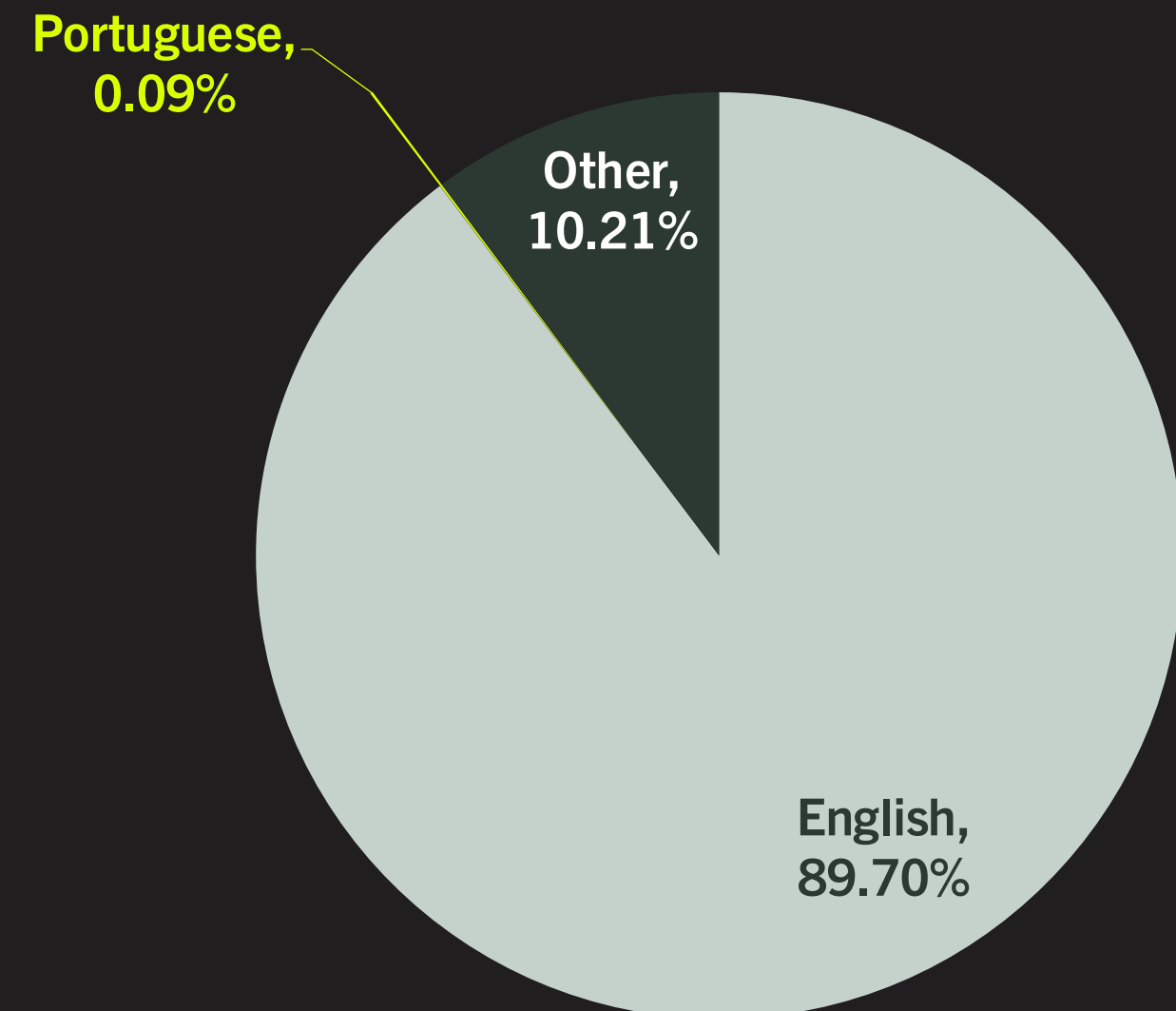
- Stanford-certified AI Specialist with 11 years of experience in AI and NLP
- Author of 10+ scientific articles and 2 book chapters; pioneer at the intersection of AI science and entrepreneurship
- CTO at Amadeus AI, a deeptech / AI Lab
- Chief Scientific Advisor of SoberanIA, the Brazilian Government's sovereign LLM project
- Executive Advisor of the Brazilian AI Association (ABRIA), the leading civil association shaping AI legislation in the Senate and Chamber of Deputies

THE CHALLENGE: THE REPRESENTATION GAP IN AI



1. **The Massive Data Disparity:** Even with 260+ million speakers globally, Portuguese is treated as a low-resource language. Standard AI training datasets are 89.7% English and only 0.09% Portuguese.
2. **The "Translation Illusion":** Global models process logic in English under the hood. Translating prompts back and forth completely erases regional nuances, local grammar, and cultural context.
3. **The Impact on Public Services:** Off-the-shelf models fail to grasp the highly complex, specific legal and administrative vocabulary required to operate in the Brazilian public sector.
4. **The Conclusion:** To solve this, we couldn't just fine-tune an existing model; we had to build our own full-stack AI pipeline from the ground up.

Language Distribution in Public Datasets





THE ENGINE: STRUCTURING A SOVEREIGN AI TASKFORCE

- **The Strategic Breakthrough:** A project cannot truly be called "sovereign" unless the public sector is leading it. Our core Gov2Gov partnership with the State of Piauí and the Federal Government (MCTI) was the fundamental enabler of our success.
- **Unlocking Closed Doors:** The government didn't just provide funding; they provided institutional weight. This opened doors to other public entities, granting us access to restricted administrative data entirely unavailable to private tech companies.
- **Attracting Top Talent:** The scale of a national sovereignty mission allowed us to assemble a multidisciplinary taskforce of 77 specialists (45% holding Masters or PhDs), including a dedicated Computational Linguistics team.

OBSTACLE 1: THE DATA WALL (CREATING A ROBUST CORPUS)

Breakthroughs in Creative Data Sourcing

When building for a low-resource language, you quickly run out of high-quality internet data. Standard web-scraping is not enough to scale a foundational model.

- **The Government Data Goldmine:** Institutional access allowed us to process 17 Terabytes of internal electronic public management documents (the SEI system). This yielded an unprecedented volume of complex, high-quality Portuguese text.
- **Capturing Authentic Dialects:** We utilized automated transcriptions of regional YouTube videos to capture local speech patterns and colloquial accents that are rarely written down.
- **Current Scale:** We successfully completed Dataset Jabuticaba (139 Billion clean tokens) and are currently scaling Dataset Lusofonia past 350+ Billion tokens (targeting 500B).

OBSTACLE 2: SOTA TRAINING WITHOUT A PLAYBOOK

Breakthroughs in our Custom Training Pipeline

We wanted to train our model using State-of-the-Art (SOTA) methods, but this knowledge is highly guarded by a few global labs. We had to execute these algorithms without support or open tutorials.

- **Continual Pre-Training (CPT):** We successfully used CPT to ingest digitized physical books and offline government materials, grounding the model in exclusive knowledge that does not exist on the public internet.
- **Supervised Fine-Tuning (SFT):** We discarded translated datasets and hand-crafted over 1 million instruction pairs deeply focused on authentic Brazilian culture, history, and local personalities.
- Used **GRPO** to ensure structural reasoning happens natively in Portuguese.
- Used **DPO** to cure the model's "identity crisis"—explicitly teaching it its sovereign origins so it wouldn't hallucinate that it was built by another lab.

OBSTACLE 3: BLIND NAVIGATION (MEANINGFUL EVALUATION)

Benchmark “Brasilidade” (“*Brazilianness*”)

20,000 meticulously crafted questions. Standard translated English benchmarks fail to capture local realities.

Evaluating 5 Core Dimensions:

- 1. Linguistics:** Normative grammar vs. colloquialisms, slang, formal/informal variations
- 2. Culture:** Brazilian history, geography, arts, and gastronomy
- 3. Applied Tasks:** Interpreting national laws, public service tasks, creative writing
- 4. Safety & Bias:** Preventing toxic discourse and respecting social sensitivities
- 5. Regional Identities:** Capturing authentic cultural subjectivity (focus on Piauí)

PROVING TRUE NATIVE COMPREHENSION

The 7 Formats of “Brasiliidade”

Preventing the model from rote-memorizing trivia or relying on surface-level translation to pass our benchmarks.

1. **Direct Q&A:** Factual local knowledge retrieval
2. **Correction & Comparison:** Identifying errors in Portuguese
3. **Classification & Annotation:** Human-style tagging
4. **Multiple Choice:** PSAT & Bar exam simulation
5. **Textual Inference:** Identifying logical contradictions
6. **Fact-Checking:** True/False with contextualized justification
7. **The Ultimate Test (Logical-Mathematical Reasoning):**
We formulate complex math word problems entirely in regional Portuguese. If an LLM can parse the cultural context of a math problem and solve it accurately natively, it proves true structural reasoning.

OBSTACLE 4: THE BLACK BOX OF SCALABLE INFERENCE



Breakthroughs in Production Engineering

Training gets all the hype, but figuring out how to do proper inference in a scalable and reliable way is a closely guarded industry secret. This is a massive hurdle for local governments with limited budgets.

- **3x Performance Optimization:** Heavy engine optimization allowed us to reduce our hardware inference requirements by 3x, drastically lowering costs and making the tech viable for large-scale public deployment.
- **Native Gov RAG System:** We built a Retrieval-Augmented Generation (RAG) system grounded directly into government databases. This definitively solves the hallucination problem for citizen-facing bots.
- **Sovereign Infrastructure:** The entire pipeline is deployed exclusively on dedicated NVIDIA GPU clusters in national Tier 3 datacenters to comply strictly with Brazil's data privacy laws (LGPD).



GOVERNANCE, BRAND & THE OPEN ECOSYSTEM

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Sovereign AI is a Strategic Necessity:

- Building LLMs for underrepresented languages is full of obstacles, but it is entirely possible if you build the right alliances—and it wouldn't be sovereign without the public sector leading.
- If we don't define our own AI ecosystem, someone else will define it for us.



BONUS: LLMS IN BRAZILIAN PORTUGUESE: A CHRONOLOGICAL SURVEY

Discover every language model in Brazilian Portuguese
From the first BERT models in 2020 to the latest finetuned LLMs



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